

India Strategy

India Strategy: Why India needs its own DeepSeek



Venugopal Garre
+65 6326 7643
venugopal.garre@bernsteinsg.com



Nikhil Arela
+91 226 842 1482
nikhil.arela@bernsteinsg.com

India can't build its AI future on borrowed models. While renting compute to power foreign LLMs may look like progress, it leaves the country dependent. The recent US restrictions on access to the latest AI model for non-citizens drive home the risk that India could find itself locked out some day, with its applications trailing those in the US and China. We had raised this topic even earlier but were muffled with noises of "LLM does not matter". But the recent development compels us to have a relook.

AI is the next 'fighter jet': Technology access has been rationed and controlled in the past, and a trend emerging in last few years is shattering the "Global" image of AI. It started with critical minerals & semiconductors equipment, moved to GPUs and now exhibits itself in the form of restricted frontier models. The recent ban on Anthropic's latest models for non-US citizens confirms that it's no longer an anomaly. As AI moves from a mere commodity to a tool of strategic importance, one thing is clear: foundational models will no longer be SaaS products, they would rather be critical resources, becoming the "fighter jets" of an era where bleeding-edge models will be guardrailed. This will only aggravate further, with deep implications for India currently trying to convince itself that applications built on foreign LLMs and earning rent on datacentres is a sound AI strategy. To be clear, we are not dismissing this approach. There is value in participating in the ecosystem through infrastructure build-out and application layers. But at the very least, India needs to be aware about the risks embedded in effectively outsourcing the core AI models.

Why doesn't India have its own large-scale LLM? It may be counterintuitive that despite being the source of vast data, which is virtually the fuel to train global models, India itself hasn't harnessed this advantage to build competitive GenAI systems at global scale. This absence of an India "DeepSeek moment" is structural rather than a thought out strategy. India's tech ecosystem has long been services-led, with sparse consumer-facing platforms in areas like search, social media or messaging—domains that typically generate the rich, organized datasets critical for training advanced AI. The lack of such data ecosystems has never necessitated the development of talent pipelines and academic depth to build foundational models. The IT services model, where low cost labor effectively fine-tunes software built by global giants, further adds to the baggage of sticking to application layer. Heads of many leading Institutions have argued that India does not need its LLMs, but can focus on AI applications. These views are more reflective of the path India has taken to stand where it is today, rather than a deliberate strategic choice.

Can India afford its AI stack at the mercy of someone else? Let's picture this: India's core intelligence layer, from enterprise software to defence and space could be powered by foreign LLMs. Enter a geopolitical disruption, and that access could be curtailed overnight, bringing critical applications to a halt. A more likely outcome is subtler but equally concerning: India operates one or two generations behind. Applications built on older models struggle to compete with global offerings. An Indian IT giant with talent at its disposal could lose out to a young US SaaS firm bootstrapped by amateur coders having access to cutting-edge models. There is, however, a path that looks more resilient—building domain-specific LLMs on proprietary data and layering AI solutions on top. The real vulnerability lies with large-scale, horizontal GenAI applications built on external models....continued on page 2

DETAILS

...continued from page 1. **Framing India's AI Policy Choices:** The solution set is not easy for policymakers. Ideas such as ringfencing AI for domestic firms (which we have given in the past) may lack practicality in a globally integrated technology landscape. That said, two strategic levers stand out. One approach is to restrict or pace access to global AI models, while channeling significant capital and talent into building India-native LLM capabilities. The other is to mandate or incentivize localization—requiring foreign firms to build and operate India-based AI stacks that are insulated from geopolitical controls. Both options are imperfect, but they frame the core policy trade-off between access and autonomy.

INDIA'S TECH DEPENDENCE

For years, India's relationship with technology has followed a familiar script. Build locally where possible, but rely on global giants where it matters. The internet era was perhaps the clearest example. While China built walls and created large scale platforms that have provided it cash flows, talent pools and data to build AI models, India stayed open, and global platforms quietly became the rails on which its digital economy runs.

But here's the catch. What started as a story of internet dependence is, in reality, much bigger. Step back and look at the full stack. From hardware and core IT components to operating systems and enterprise software, India sits downstream across almost every layer. [Exhibit 1](#) and [Exhibit 2](#) make this stark. This is not just about apps or platforms; it is about systemic dependence. And overwhelmingly, that dependence points in one direction: the US. There have been attempts by Indian companies to break through in pockets, to build credible vertical platforms, but most have struggled to scale, out-competed by global incumbents with deeper moats, better products, and capital at scale.

Which brings us to today, and to AI. If history is any guide, the path forward seems obvious. Why fight it? Why not allow the same playbook to repeat, with global companies building, owning, and monetizing India's AI layer while domestic players participate on the margins? And yet, something feels different this time. AI is not just another application layer. It is emerging as the underlying rail for everything from enterprise workflows to consumer interactions. Importantly, it is still early. Unlike past technology cycles where India entered late, this is a moment where the contours are still being shaped. That is what is triggering the current debate around a sovereign AI stack. Not because India has suddenly discovered technological self-reliance, but because the cost of missing this wave could be far higher. Waiting 20–30 years, as India effectively did in earlier tech cycles, could entrench dependence in ways that are far harder to reverse, locking in value capture outside the country for decades.

But here lies the dilemma. Recognizing the importance of sovereignty is one thing; executing on it is another. The policy levers are limited, the competitive gap is large, and the alternatives outlined in our framework are far from straightforward. India may not dominate the core AI stack, but the choices it makes now will determine where value pools emerge, who captures them, and whether the next decade of AI-led growth deepens dependence or begins to rebalance it. The debate on LLMs, therefore, is not just a technology discussion. It is an early signal of how India intends to position itself in the AI economy. And this time, the decision cannot be deferred.

We believe India's real opportunity lies in owning and leveraging its rich, domain-specific datasets, especially across sectors like Industrials and healthcare. These datasets can become the foundation for building defensible, smaller, specialized LLMs by Indian technology companies, on top of which a wide range of applications can be developed. Importantly, these applications need not be confined to the digital world. They can extend into the physical economy as well, powering use cases such as training industrial robots, including next-generation humanoid systems where perhaps the Industrial context specific data set is more critical than the use of the latest Gen AI model and where older gen AI models may also work. The opportunity is, therefore, not just in software, but in shaping how AI integrates with real-world assets and workflows.

The core idea is simple but powerful: retain and control access to high-quality, hard-to-replicate vertical data. By limiting unrestricted access to such datasets for global platforms, India can create pockets of defensibility where domestic players can build meaningful capabilities. Over time, these can evolve into scalable solutions within India, and eventually be exported as global offerings. Realistically, this path may not produce trillion-dollar tech giants in the near term. But that is not the only metric of success. Even a shift toward building credible, globally relevant platforms anchored in proprietary data would expand India's share of value creation beyond IT services, and gradually reposition it within the global tech ecosystem. More importantly, it sets the foundation for the future. As talent and capital begin to recognize the viability of an India-led tech stack, it could trigger a virtuous cycle—one where the next generation of entrepreneurs builds not just for India, but from India.

EXHIBIT 1: India's tech dependence

Area	Main companies India depends on	Country of origin	Indian providers / initiatives
Core compute: CPUs, GPUs, chipsets			
PC/server CPUs	Intel, AMD	Intel: US; AMD: US	CDAC's ARM designs for niche HPC; no mass CPU yet
Mobile SoCs	Qualcomm, MediaTek, Apple (A-series), Samsung Exynos	US; Taiwan; US; South Korea	Jio/Reliance design efforts (limited), none at scale
AI NPU's / accelerators	NVIDIA, AMD, Google TPU (cloud), Intel Gaudi	US; US; US; US	None in production; early RISC-V and fabless efforts
GPUs (datacenter/client)	NVIDIA, AMD	US; US	None meaningful; India is a GPU consumer only
Memory (DRAM, NAND)	Samsung, SK Hynix, Micron	South Korea; South Korea; US	instagram
Storage controllers, SSDs	Samsung, Western Digital, Seagate, Kioxia	KR; US; US; Japan	No large domestic DRAM/NAND makers
Operating systems (PC, mobile, server)			
Desktop / laptop OS	Microsoft Windows (dominant), Apple macOS, desktop Linux	US; US; global OSS	Only assembly/integration by Indian system integrators
Mobile OS	Google Android (incl. OEM skins), Apple iOS	US; US	Maya OS (Govt Linux-based for defence)facebook
Server OS	Linux distros (Red Hat, Ubuntu, etc.), Windows Server	US/UK; US; others	BharatOS / homegrown Android forks (small share)
Desktop productivity, content creation, key software			
Office suites	Microsoft 365/Office	US	Local Linux customizations in govt/PSUs
PDF tools	Adobe Acrobat	US	Zoho Workplace (Zoho, India); LibreOffice (OSS)
Creative suite (photo, video, design)	Adobe Creative Cloud, Corel, Autodesk	US; Canada; US	Small Indian PDF tools; none at Adobe's scale
IDEs & dev tools	Microsoft (VS, VS Code), JetBrains, open-source tools	US; Czech/Netherlands	Inkscape, GIMP (OSS); few Indian pro-grade suites
Browsers	Google Chrome, Apple Safari, Microsoft Edge, Firefox	US; US; US; US	Indian Chromium forks possible; no large consumer brand
Enterprise applications: ERP, accounting, CRM			
Large-enterprise ERP	SAP, Oracle, Microsoft Dynamics, Infor, Sage	Germany; US; US; US; UK	Ramco Systems, Tally Solutions, Focus Softnet, Bsquare Pothera ERP
SME accounting / ERP	Tally (very dominant), Zoho Books, QuickBooks	India; India; US	Tally Solutions (India), Zoho (India)
CRM / SaaS business apps	Salesforce, Microsoft, HubSpot, SAP, Oracle	US; US; US; Germany; US	Zoho CRM (India), Freshworks (Freshdesk, etc.), LeadSquared
HR / payroll	SAP SuccessFactors, Oracle HCM, Workday	Germany; US; US	Darwinbox, Keka, GreytHR (India)
PCs, laptops, and other client hardware			
Laptops / PCs	HP, Dell, Lenovo, Acer, Asus, Apple	US; US; China; Taiwan; Taiwan; US	HCL (legacy PCs), Reliance, Lava, Dixon assembling for brands
PC components	Motherboards, GPUs, RAM, SSDs from global brands (ASUS, Gigabyte, MSI, etc.)	Taiwan; Taiwan; Taiwan	No major Indian component brands; mostly import-dependent
Monitors	Samsung, LG, Dell, HP	KR; KR; US; US	A few Indian labels (e.g. Intex, Zebronics) using imported panels

MediaTek, Samsung, SK Hynix, Micron and Kioxia are covered by Mark Li; Microsoft, Adobe, SAP, Oracle, Salesforce, HubSpot and Workday are covered by Mark Moerdler; Apple, Western Digital, Seagate, HP and Dell are covered by Mark Newman; Alphabet (owns Android, Chrome) is covered by Mark Shmulik; Sage is covered by Richard Nguyen; Intel, AMD, Qualcomm and Nvidia are covered by Stacy Rasgon

Lenovo, Acer, Asus, Gigabyte, MSI, LG Electronics, Reliance, Ramco Systems, HCL Tech and Freshworks are not covered

Remaining companies are unlisted

Source: Company Data, Bernstein Analysis

EXHIBIT 2: India's tech dependence

Area	Main companies India depends on	Country of origin	Indian providers / initiatives
Smartphones, feature phones, and mobile hardware			
Smartphones (brands)	Samsung, Xiaomi, Vivo, Oppo, Apple, OnePlus, Motorola	KR; China; China; China; US; China; US	Lava, Micromax (smaller), JioPhone (Reliance), Karbonn
ODM / manufacturing	Chinese/Taiwanese ODMs; Foxconn, Wistron, Pegatron, etc.	Taiwan; Taiwan; Taiwan	Dixon, local EMS units under PLI schemes
Modems / RF chips	Qualcomm, MediaTek, UNISOC	US; Taiwan; China	No domestic RF baseband vendor
Displays & batteries	Samsung, LG, BOE, CATL, others	KR; KR; China; China	Limited local battery assembly, cells largely imported
Telecom networks and equipment			
Mobile RAN & core	Ericsson, Nokia, Huawei (historically), ZTE, Cisco	Sweden; Finland; China; China; US	Tejas Networks, HFCL, ITI, C-DOT-aligned vendors
Transmission & fiber	Nokia, Ericsson, Huawei; global fiber makers	Finland; Sweden; China	HFCL, Sterlite Technologies (STL) for fiber and some gear
Tower infra	American Tower, global OEMs for gear	US, others	Indus Towers (India), Bharti Infratel (now merged)
Routers/switches (IP)	Cisco, Juniper, Huawei, Nokia	US; US; China; Finland	Some L2/L3 switches by Indian OEMs; not yet at Tier-1 scale
Cloud, hyperscale, and developer platforms			
Public cloud IaaS/PaaS	AWS, Microsoft Azure, Google Cloud	US	Niche Indian DC/cloud providers (CtrlS, ESDS, Krutrim)
SaaS productivity	Microsoft 365, Google Workspace, Zoom, Slack	US	Zoho suite (India), Flock (India)
Code hosting	GitHub, GitLab,	US	Few Indian Git alternatives; minor adoption
Social media, messaging, consumer internet			
Social media	Meta (Facebook, Instagram), X (Twitter), LinkedIn	US	ShareChat, Koo, Josh (short video)
Video streaming	YouTube, Netflix, Amazon Prime Video, Disney+ Hotstar	US	JioCinema, SonyLIV, Zee5
Messaging apps	WhatsApp, Telegram, Signal, iMessage	US; UAE/Cyprus; US; US	Hike (pivoted), JioChat; limited scale now
E-commerce	Amazon, Walmart/Flipkart	US	Reliance JioMart, Tata Neu, Snapdeal
Payments / UPI	Google Pay, PhonePe (Walmart-backed), Paytm	US; US; India	Paytm (India), many Indian UPI apps
Peripherals and other PC/mobile parts			
Keyboards, mice	Logitech, HP, Dell, local OEM imports	Switzerland; US; US	Zebronics, Intex, iBall (brand/OEM, not deep IP)
Printers, scanners	HP, Canon, Epson, Brother	US; Japan; Japan; Japan	Very limited Indian printer IP; mostly imports
Power adapters, chargers	OEMs from China/Taiwan	China; Taiwan	Many small Indian brands; core designs imported
PC cabinets, PSUs	Cooler Master, Corsair, Antec, OEM imports	Taiwan; US; US	Local brands (e.g. Artis); still largely imported components

Xiaomi is covered by Eunice Lee; Netflix is covered by Laurent Yoon; American Tower is covered by Madison Rezaei; Samsung is covered by Mark Li; Microsoft (owns Github) and Salesforce (owns Slack) are covered by Mark Moerdler; Apple, HP and Dell are covered by Mark Newman; Amazon, Alphabet and Meta are covered by Mark Shmulik; CATL is covered by Neil Beveridge; Zoom and GitLab are covered by Peter Weed; Paytm is covered by Pranav Gundlapalle; Ericsson and Nokia are covered by Ulrich Rathe; Walmart is covered by Zhihan Ma

Hon hai, Wistron, Pegatron, BOE tech, Lenovo, Reliance, HFCL, Sterlite Tech, Indus Towers, Bharti Airtel, Ola (owns Krutrim), Zee, ZTE, Cisco, Juniper, Logitech, Canon, Epson, Brother Industries, Corsair, Antec and TCS (owns Tejas Networks) are not covered

Rest of the companies are unlisted

Source: Company Data, Bernstein Analysis

INDIA'S AI APPROACH SO FAR

Much like the period following the computer revolution, Indian technology leaders may be at risk of falling into a similar structural challenge, this time with potentially far greater consequences. Historically, India's technology industry was built on a services-led model. While global firms developed the core platforms that powered digital ecosystems, Indian companies excelled at maintaining, customizing, and extending these platforms through incremental layers of code and services. This model proved highly effective in an era defined by large software systems with relatively modest customization needs.

However, the dynamics with AI appear fundamentally different. As foundational models become increasingly powerful, the value may shift away from incremental application-layer work. Applications built on top of these models could become more standardized and widely accessible, reducing the differentiation that service-led models once provided. An additional layer of complexity arises from potential dependencies on foundational models developed and controlled outside India. If access to these models were to be constrained, for instance by policy decisions in countries where they are developed, it could introduce strategic vulnerabilities for companies that rely heavily on them.

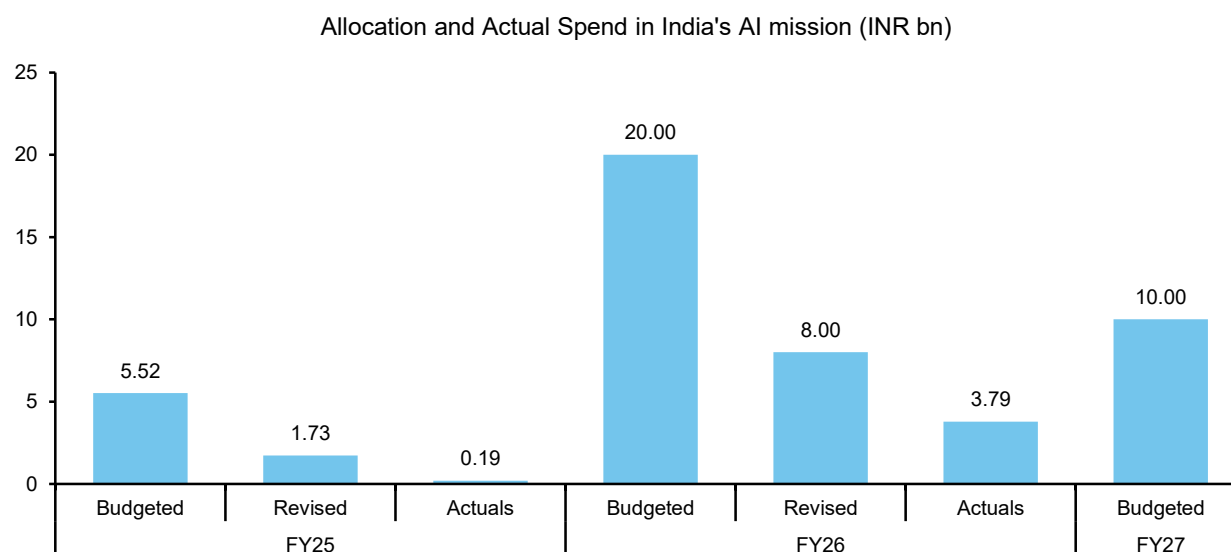
TOO LUMPY, TOO LITTLE

While this approach can be viewed as a cautious response shaped by past experience, it is also important to acknowledge another reality. India did see early momentum around foundational models, with several announcements that later lost traction or evolved into different directions. Some Indian companies initially outlined ambitions to build large, multi-billion-dollar LLMs, but over time these plans were either scaled back or reoriented.

At the policy level, the India AI Mission, launched in 2024 with an outlay of approximately INR 104 billion, or about USD 1.2 billion spread over five years, identified compute, data, research, and foundational models as key pillars. Since then, participation has become more focused, and among the original set of players, Sarvam currently appears to be one of the most committed efforts toward building a sovereign LLM stack.

At the same time, funding trends have been uneven. Allocations toward AI saw a notable reduction in the revised FY2025 budget in 2026, highlighting some volatility in execution. More broadly, the overall financial commitment, while meaningful, remains modest when viewed in a global context. For instance, reported public investments in AI in China are significantly larger, running into tens of billions of dollars annually.

EXHIBIT 3: Too lumpy, too little: India's AI strategy so far has been that of trying to do a lot from less resources



Actuals for FY26 are till 9th February, 2026 when government responded to a question on India's AI mission in the Rajya Sabha
Source: Budget Documents, Government Response in Rajya Sabha, Bernstein Analysis

DEFOCUSED, THINLY SPREAD

Beyond funding, the broader strategic approach also appears widely distributed across multiple priorities, which may make it challenging to achieve meaningful depth in any one area. As noted earlier, India's AI Mission was framed with an ambitious scope, spanning compute, data, research, hardware, applications, and foundational models. While comprehensive in intent, such a wide canvas can make focused execution more difficult.

Within the overall allocation of approximately USD 1.2 billion over five years, only a portion is directed toward foundational models. A little over USD 220 million, or under 20 percent, has been earmarked for this area. A similar amount has been set aside to support startups over the same period, while a significantly larger share, about 44 percent, is directed toward expanding access to compute capacity, including through imports.

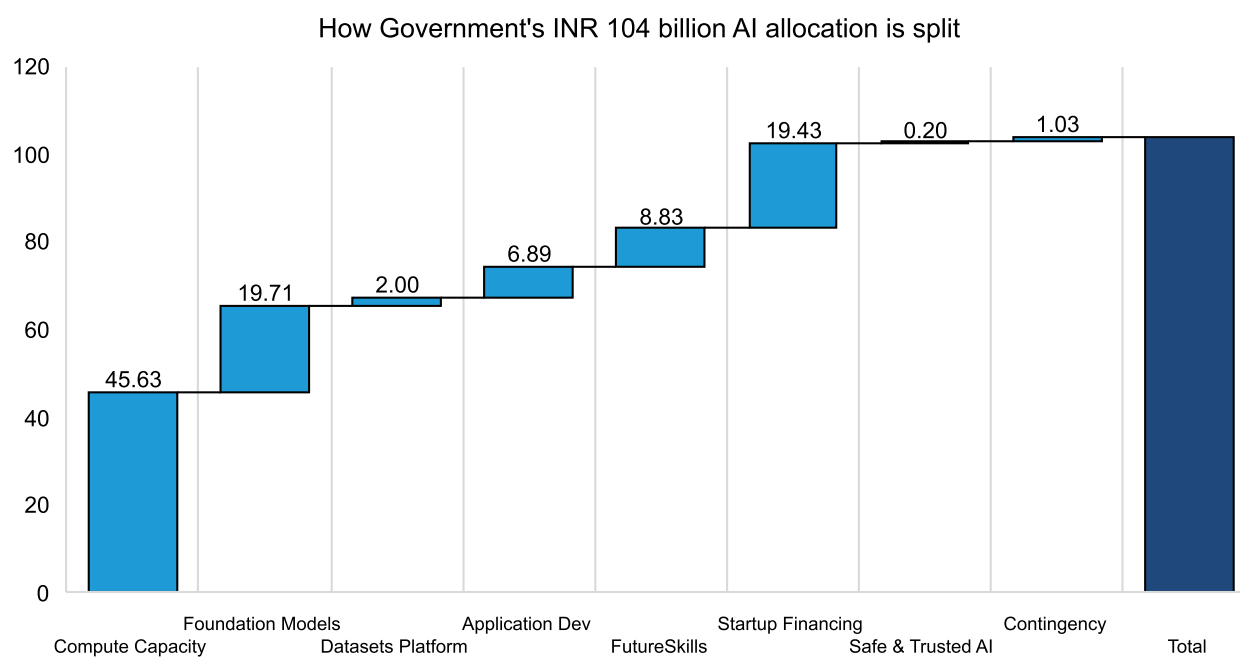
In practice, there is often a gap between initial ambition and eventual outcomes. However, in this case, the challenge may begin at the level of prioritization. The current approach reflects an effort to build capabilities across multiple layers of the AI stack at once. While this ensures broad participation, it may also dilute the ability to build leadership in any single segment.

This contrasts with the more sequential approaches observed in other major AI ecosystems. In the United States, early momentum was anchored in access to compute and the strength of private capital, which enabled rapid advances in model performance alongside supporting infrastructure. China, on the other hand, placed a strong emphasis on model efficiency, while also investing in semiconductor capabilities in response to external constraints. Over time, both ecosystems expanded their focus across the stack, but this followed an initial phase of concentrated investment in specific strengths.

A related consideration is the scale of investment required. Building capabilities across multiple areas of AI simultaneously demands significant capital and institutional depth. In the absence of this, spreading resources too broadly can limit effectiveness, particularly in areas where domestic capabilities are still evolving. In such cases, a portion of the investment may flow toward procuring external technologies, which can reduce the impact on local capability building.

The other question is: should the Government be even involved? The involvement if any should be on the policy front - like "the great firewall" in China as long as there are identified Indian companies willing to participate in the LLM race. If there aren't any - and everyone wants to build applications then even a Government su

EXHIBIT 4: Too defocused, thinly spread: Even with lesser and lumpy allocations, the focus is all over the place, creating no single area with meaningful advantage for India



Source: Parliamentary Response by the Government, Bernstein Analysis

THE CRACKS IN THE AI 'GLOBALISATION'

Technology is often cited as a great leveller, but that misses an important caveat hidden between the lines - it is a great leveller given everyone has equal adoption, and more importantly, equal access to it. In the absence of these, particularly the latter, it becomes the catalyst for a widening gap that continuously compounds with time. Technology, particularly in sectors such as defence, nuclear, and chemicals, has long been treated as a strategic sovereign asset. A new theme has joined these ranks over the last 4-5 years - the theme of Artificial Intelligence, as it has been reframed from a general purpose technology into a critical resource giving nations an edge over others. Governments have increasingly used classic export control tools to lock down key inputs like high-end GPUs, most visible in the successive US rounds of restrictions on exporting advanced semiconductor technology to China ([Exhibit 5](#)). At the same time, national strategies increasingly have started to talk about "sovereign AI", where control over not just data, but even compute and foundational models is treated as paramount for gaining (or retaining) future economic and geopolitical power.

AI has often been framed as a force for global progress. However, early signs suggest that this vision of a seamless, globally accessible AI ecosystem is not evolving in practice. Initial measures focused on restricting access to advanced chip fabrication technologies and Electronic Design Automation tools. This was followed by limitations on the export of high-end GPUs. More recently, access to certain advanced AI models has also become more selective, reflecting a broader shift in how countries are approaching control over critical technologies, while adding LLMs as the newest entry to the list of restricted technologies. The trend only points one way → we will increasingly see more section 232s and 301s in the post-AI world.

EXHIBIT 5: US limiting actions on AI and related equipment on the world (primarily China)

Month	US Limiting Action	
Oct-22	Semiconduction Manufacturing Equipment and Advance logic chips added to Commerce Control List (CCL)	
Oct-23	China entities involved in advanced computing added to Entity List (EL)	
Oct-23	Worldwide license requirements imposed for usage on behalf of entities either headquartered in, or with parent companies headquartered in China	
Oct-23	License requirements for chips designed for China - below October 2022 thresholds	
Dec-24	Expanded the Foreign Direct Product Rule to apply to SME and Chips	
Dec-24	Added 140 Chinese and China-linked entities to EL	
Jan-25	AI diffusion rule proposed, dividing the countries into three tiers based on relations with US. Tier to be instrumental in determining what can be exported	
Mar-25	Added 42 China entities to the EL, rescinded AI diffusion rule	
May-25	Issued guidance that any use of Huawei's Ascend AI chips violates U.S. export controls.	
Sep-25	Removed named PRC facilities of Samsung and SK Hynix from the Validated End User program. Added 23 China entities to the EL.	
Jun-26	US issues export control directive to Anthropic to suspend access to Fable 5 and Mythos 5 for all foreign nationals, including non-citizen employees inside the US	
US Chip Restrictions		
Oct-22	Controls on A100 and H100 chips	Modified China chips announced by Nvidia: A800 and H800
Nov-23	Restrictions on A800 and H800 chips	New chips announced for China: H20, L20 and L2
May-25	Initial controls on H20	Nvidia announces efforts to develop B30 and B40 for China
Jul-25	Export license for H20 Sales to China granted	
Sep-25	Reports of BIS consultations with Nvidia regarding the B30 and possibly the B40	Cyberspace Administration of China effectively "blacklisted" the B40

Nvidia is covered by Stacy Rasgon

The US Entity List is a trade restriction published by the US Department of Commerce's Bureau of Industry and Security (BIS). It identifies foreign individuals, companies, and organizations believed to be involved in activities contrary to US national security or foreign policy interests, subjecting them to strict export license requirements

The Foreign Direct Product Rule (FDPR) is a strict U.S. export control that extends American regulatory jurisdiction to products manufactured entirely outside the United States

Source: Congress.gov, Bernstein Analysis

EXHIBIT 6: **Alternate platforms from China for US technology**

Huawei	US/Global
Kirin 9000S	Qualcomm Snapdragon
Kirin X90	Intel Core processor
Kunpeng 920	x86-based server CPU
Ascend 910C	Nvidia H100 & H20
CloudMatrix 384	Nvidia GB200 NVL72
CANN	Nvidia CUDA
EulerOS	Other Linux-based server OS
MindSpore	PyTorch, TensorFlow
HarmonyOS Next	Android mobile OS
HarmonyOS PC	Windows OS
MetaERP	Oracle, SAP
GaussDB	Oracle SQL Developer

Huawei is not listed. Nvidia, Qualcomm and Intel Corp are covered by Stacy Rasgon. Alphabet (owns Android) is covered by Mark Shmulik. Microsoft (owns Windows) and Oracle are covered by Mark Moerdler.

Source: Brookings, Bernstein Analysis

LESSONS FROM OTHER INDUSTRIES - AND WHY AI WILL FOLLOW

Even though China is the current focal point, this largely reflects the reality that it is the strongest challenger to Western dominance in the global AI race. It does not insulate India from future potential actions, and in fact, India can very well be caught in the cross-fire of the global sovereign AI conflict. It isn't surprising too, since throughout history, India has been no stranger to west-led sanctions - and they have chased it across fields like Nuclear, Space tech and defence ([Exhibit 7](#)).

The trend has already begun in AI, and it's only a matter of time when AI is so deeply entrenched in all systems that it moves from a corporate or academic productivity enhancer to a major tool that nations and their infra depends on. And that will be the time lack of sovereign models will hurt the most. Mere application layer presence, however wide, will not work. Imagine your entire layer working on top of foundational models whose access can be turned on or off like a tap by another nation thousands of miles away. Deep capabilities become a necessity, not a luxury.

The IT services parallel to AI is promising but ultimately misleading: while IT services in India exported Indian cheap labor and applied it to foreign platforms, the application layer of AI will rather export value creation that is powered by the very foundational models west has built. So while there was little alternative to the former in the west, the latter can have startups springing to life by previously non-tech savvy people or "vibe coders", spearheading the west into the "services powered by AI" era.

EXHIBIT 7: **Export controls and restrictions placed on India**

Field	Period	Incident	Subsequent Technology/Items Restrictions	Impact on India
Nuclear	1974	India's nuclear tests triggered concerns over proliferation	Advanced nuclear reactors, fuel-cycle technology, heavy water, reprocessing and enrichment tech placed under tight export controls to India	Slowed India's access to latest reactor designs and fuel, constrained nuclear power growth. India doubled down on domestic ecosystem and pursued 3 stage thorium-based program
Space	1990s	Concern that India's space launch vehicle work could aid ballistic missile programs; MTCR rules led to restrictions on missile-relevant tech transfer	Sanctions on Indian Space Research Organisation (ISRO) and related entities; denial of dual-use items like high-end electronics, materials, and some satellite/launch tech	Restricted access to some high-performance components, delayed a few commercial collaborations and increased cost/time to source equipment from non-US suppliers. Also instrumental behind India's indigenisation and launch capabilities today being largely domestic
Defence/ Nuclear	1998	Pokhran-II triggered mandatory US sanctions and similar measures by other OECD partners	Suspension of foreign aid and most military sales, denial of dual-use exports for nuclear/missile reasons, opposition to non-basic-human-needs multilateral lending. Japan halted new yen loans	Short-term macro and financial pressure and tighter access to high-end dual-use equipment for industry and research
Semi-conductors	2018	Concerns over military applications of AI, advanced chips, quantum, drones and additive manufacturing; focus more on China/Russia but controls are country-agnostic and apply to non-NATO countries	Stricter licensing on advanced semiconductor tools, quantum devices, AI accelerator chips, some high-precision machine tools and additive manufacturing equipment.	Friction in acquiring cutting-edge fab equipment and some high-end AI hardware under blanket dual-use controls, even with deepening strategic partnership with the US/EU

Source: Bernstein Analysis

WHAT THE AI FUTURE WILL LOOK LIKE - THE LIKELY AND THE EXTREME CASE

In the most plausible trajectory, AI is likely to settle into a world of stratified access. Advanced nations or blocs/alliances will have their own powerful foundational models, but will reserve the bleeding-edge capabilities for domestic use in defence, intelligence and critical infrastructure. Exportable versions will be older or fine-tuned to be less lethal, as is the case with defense. In practice, that will restrict emerging markets like India to N-1 or N-2 models. Over time, the effects of that artificially created gap will compound - whoever controls the most powerful foundational models rules military planning, owns the best intelligence and shapes scientific discoveries in the new age.

The extreme scenario is even harsher. AI models become strategic exports - reserved for only the elite and their allies. Access can be turned off not just for the governments, but also companies and researchers in “unfriendly” countries or countries that simply lose favor with a major AI power. India could one day wake up and find that the most powerful applications in finance or military are no longer working, and the APIs and weights they rely on have vanished overnight. For as much talent as Indian IT services companies may boast of, an upstart business bootstrapped by vibe coders elsewhere that have access to cutting edge models may ultimately give them a tough competition, taking away all the labor moat.

India cannot afford to operate on someone else's trapdoor. If foundational models represent the new strategic high ground, this demands a clear pivot in India's technology ambitions. While India may not replicate China's state-led behemoths or field multiple firms with the balance sheets to deploy tens of billions in R&D, it doesn't need to. A portfolio of “imperfect” but pragmatic strategies—leveraging its talent depth, open ecosystems, and frugal innovation can still carve out a meaningful edge in what is fast becoming a contested global AI landscape.

India's unique data reservoir and digital infra is a big leverage. Government mandates on critical sector models (finance, health, defence, etc) using data to be built and trained inhouse can be a starting step. India's far more access to global fabs and GPUs compared to China enables it to build domestic training capacity, and pair horizontal LLMs with deep vertical champions in domains where India is structurally strong - or which are critical for national security. With smaller steps taken, India may even realize the goal of a general purpose LLM some day. This is no longer a side hustle, but a critical component in nation building. India's own “DeepSeek” has stopped being a luxury; it is now a means to ensure that its brightest engineers do not suddenly discover one day that they are merely petty citizens of a transformed technology empire, ruled by someone else.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

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